

Spinning Donut

same one idea as the pyramid take a dot in 3D, do math on it, it appears on 2D screen, do that for every dot, 60 times a second, your eye sees a spinning donut

what is a donut in math

a donut shape is called a torus to describe every point on its surface we only need two angles

- $i \rightarrow$ sweeps around the tube cross section, like going around a circle drawn on the tube itself
- $j \rightarrow$ sweeps that circle around the big ring, like walking around the whole donut

together i and j can reach every single point on the donut surface, that is the whole torus formula, once you have a surface point from i and j , everything after that is identical to the pyramid, rotation matrices using angles A and B spin the donut, perspective divide using $invZ$ flattens it onto the screen, zbuffer makes sure close parts hide far parts

lighting the new thing donut has that pyramid did not

pyramid used different ascii chars per face as fake shading. donut does real lighting math, N is the dot product of the surface normal with the light direction. dot product is just multiply each component and add them up, that result telling us the angle between the surface and the light

- surface facing directly toward light if N is high then bright char like `#` or `@`
- surface facing away if N is low or negative then dark char like `.` or `,`

Using these[`.,_~:;=!*#$$@`] these are chars going from almost empty to almost solid. N picks which one, this is called lambertian lighting the simplest real lighting model in computer graphics